END SEMESTER EXAMINATION JUNE 2025

Roll No.: .....

Name of the Program: B. Tech (CSE) **Semester: IV**

Name of the Course: Fundamental of Cyber Security **Course Code: TCS 495**
Time: 3 Hours **Maximum Marks: 100**

Note:

(i) All the questions are compulsory.
(ii) Answer any **two** sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1	**Attempt any two parts only** **10X2**	
(a)	Why there is need of cyber security, elaborate with five security breach incidents? How three pillars of cyber security protect systems and organization.	CO1
(b)	Describe following: i) Types of hackers ii) Types of attacks iii) Penetration testing lifecycle	CO1
(c)	Define the goal of zero trust model. Elaborate how the steps of ZTM protects the system and its resources?	CO1
Q2	**Attempt any two parts only** **10X2**	
(a)	Define shell in Linux operating system. In layered approach model what is the layer of shell in OS. Explain the functionality of system calls.	CO4
(b)	Which command used to check permission given to a file and directory. Explain with symbolic and numeric representation of permission. i) Write the syntax of grep, head, touch, vi editor and sudo commands. ii) Suppose an attacker want to run a script file a certain specified time without human intervention and store that scanned data in a file. Describe the fields and property of commands used in above case.	CO4
(c)	Write the program for the following cases: i) Write a shell script to store IP address, ports, OS system versions etc. of system in a file. ii) Write a program in python to store a website links and directories in a file.	CO4
Q3	**Attempt any two parts only** **10X2**	
(a)	Explain TCP/IP protocol model. And how each layers of protocols helps attacks and security profession for attacks and investigation.	CO3
(b)	Explain Virtualization and types of hypervisors. In how many ways network connection can be made on a host machine and among different VMs.	CO3
(c)	In a network an attacker wants to send payload to a victim. Identify the tool used by the attacker in this case and describe all features of that tool.	CO3
Q4	**Attempt any two parts only** **10X2**	
(a)	Differentiates following with examples:	

	i) Create a form page in HTML with JavaScript validation contains three fields name, mobile_number and email_id. ii) Write a code in PHP to connect with MySQL database.	CO6
(b)	What is client and server. How this model works explains with diagram? List all the vulnerabilities that may occurs in this model.	CO4
(c)	Describe following: i) An attacker from client system redirected the URL in his system by changing the web scrips. Which kind of attacked performed by the attacker. Explain. ii) By what techniques an attacker can perform an attack from client system to server database. Illustrate with examples.	CO4
Q5	**Attempt any two parts only** **10X2**	
(a)	Explain following with use case: i) VPN ii) SSH iii) SSL iv) DDoS	CO5
(b)	When executing a program segmentation fault error occurred. If it will not have addressed by the programmer, then what kind of attack may be performed by the attacker. Identify the attack and describe it.	CO6
(c)	Describe the following with examples: i) IP Spoofing attack ii) Birthday Attack iii) Ransomware Attack iv) Eavesdropping Attack	CO5

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